Meng'en Qin

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Summary

Hi, I'm Meng'en Qin, a statistics graduate from Henan University. My research journey began at the Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms, where I explored interpretable and theory-guided visual representation learning. More recently, I have been investigating the internal scientific mechanisms of MLLMs/LLMs, particularly their reasoning and hallucination behavior. Going forward, I hope to develop scientific approaches to understand AI and use these insights to build more interpretable, robust and reliable models.

Education

School of Mathematics and Statistics, Henan University — *Kaifeng, China*

B.S. in Statistics — *Sep 2021 - Jun 2025*

- Faculty Ranking: top 8% and GPA: 87/100

Research Experience

Shenzhen University of Advanced Technology — *Shenzhen, China*

Research Assistant — *Sep 2025 - Sep 2026*

- scientific mechanism investigation of MLLMs/LLMs (working with Prof. Ruize Han, Prof. Youlu Xing and Distinguished Prof. Song Wang)

Henan Engineering Research Center for Artificial Intelligence Theory and Algorithms — *Henan, China*

Research Intern — *Nov 2023 - Jan 2025*

- interpretable and theory-guided visual representation learning (work with Prof. Xiaohui Yang)

Honors & Awards

Nov 2023	**Provincial First Prize**, China Undergraduate Mathematical Contest in Modeling	*Henan, China*
Aug 2023	**National Second Prize (Top 0.5%)**, National College Student Statistical Modeling Competition	*China*
Aug 2023	**National Third Prize**, National College Students Social Practice and Science Contest on Energy Saving & Emission Reduction	*China*

Papers

MLLMs Hallucinate when Information Distribution Drifts in Synergy Heads — *arXiv:2609.09206*

Meng'en Qin, Junye Chen, Jucheng Liu, Yinchen Liu, Youlu Xing, Song Wang, Ruize Han† — *2026*

- We identify visual-language distribution within synergy heads as a causally intervenable factor that can influence hallucination behavior, and propose a theoretically grounded calibration strategy to steer the representation geometry toward the equilibrium direction.

Adaptive Compressed Residual: Redundancy-Aware Mechanism for Compact Multimedia Representation Learning — *revision resubmitted to TMM*

Meng'en Qin*, Yu Song*, Jing Fang, Deyu Meng, Xiaohui Yang†, Lili Yang† — *2026*

- We revisit residual connections from a theoretical and redundancy-aware perspective and propose adaptive sparse compression on the skip pathway, enabling selective suppression of task-irrelevant components while preserving discriminative information.

Training-Adaptive Convolutional Sparse Coding via Information Bottleneck for Robust Visual Representation — *arXiv:2609.19122*

Meng'en Qin, Yinchen Liu, Mingxuan Cui, Youlu Xing† — *2026*

- We establish an explicit connection between the sparse representation and information bottleneck principle and provide an interpretable lens to learn compact and sufficient visual representations, improving model robustness under corrupted inputs.

A^3-FPN: Asymptotic Content-Aware Pyramid Attention Network for Dense Visual Prediction — *Pattern Recognition*

Meng'en Qin, Yu Song, Quanling Zhao, Xiaodong Yang, Yingtao Che, Xiaohui Yang† — *2026*

- We revisit the existing multi-scale visual learning networks and propose a theory-guided Asymptotic Content-Aware Pyramid Attention Network to augment multi-scale feature representation for dense visual tasks.

StomaD^2: An All-in-One System for Intelligent Stomatal Phenotype Analysis via Diffusion-Based Restoration Detection Network — *arXiv:2604.18632*

Quanling Zhao*, Meng'en Qin*, Yanfeng Sun, Yuan Miao†, Xiaohui Yang† — *2025*

- We propose a unified framework that integrates diffusion-assisted enhancement, context-aware detection, and quantitative stomatal trait extraction within a single pipeline for noninvasive field leaves.

Significance and Stability Analysis of Genotype-Environment Interaction using GxEStat *submitted to Journal of Applied Statistics*

Meng'en Qin, Zhe Li, Hui Huang, Xihong Liu† *2026*

- We propose a statistical analysis framework for multi-environment breeding trials, and extend classical stability analysis by combining single-genotype group regression with multi-genotype matrix factorization approaches.

OS-MSWGBM: Intelligent Analysis of Organic Synthesis Based on Multiscale Subtraction Weighted Network and LightGBM *MATCH Commun. Math. Comput. Chem.*

Lanfeng Wang*, Yanhui Guo*, Zelin Zhang, Meng'en Qin, Zixin Li, Xiaoli Sun, Xiaohui Yang† *2025*

OCS-TGBM: Intelligent Analysis of Organic Chemical Synthesis Based on Topological Data Analysis and LightGBM *MATCH Commun. Math. Comput. Chem.*

Yanhui Guo*, Lichao Peng*, Zixin Li, Meng'en Qin, Xue Jiao, Yun Chai and Xiaohui Yang† *2024*

Patents

Information Intelligent Compensation and Compression Based on Progressive Fusion of Spatial and Channel Features *CN119131792B*

Lichao Peng, Meng'en Qin, Xiaohui Yang, Chen Miao, Yanfeng Sun, Han Qiao *2025*

Coal and Gangue Intelligent Recognition Based on Selective State Space Equation *CN119048740A*

Xiaohui Yang, Meng'en Qin, Yanni Zhang, Zheng Ge, Yunfei Bai *2024*

Intelligent Analysis Method of Stomatal Phenotype of Crop Non-destructive Leaves Based on HDIoU *CN118691972A*

Lichao Peng, Meng'en Qin, Xiaohui Yang, Chen Miao, Yanfeng Sun *2024*

Multimodal Data Integration Based on Convolutional Sparse Coding and Optimal Transport *CN117763499B*

Xiaohui Yang, Jingjing Li, Yuan Feng, Lichao Peng, Yu Song, Meng'en Qin *2023*

Extracurricular Activities

Stellar Corps, Young Sprout Association *Yuqing Middle School, Guizhou*

Vice Captain and Teaching Volunteer *Jan 2023 - Dec 2024*

- I joined the Young Sprout Association of Henan University and served as vice captain of Stellar Corps (for high school students). We collaborated with Shanghai Lingqing Philanthropy Development Center and co-launched a long-term volunteer project with Yuqing Middle School in Baini Town, Guizhou Province, China. We spent most vacations on volunteer teaching activities, student mentoring, and course design.

Debate Team of School of Mathematics and Statistics *Henan University*

Fourth Debater *Oct 2021 - Dec 2022*

- I served as the fourth debater on the School of Mathematics and Statistics debate team. As the closing speaker, I delivered final rebuttals, synthesized arguments, and crafted impactful conclusions.